

A developer who digs into root causes and enjoys solving difficult problems

Backend Engineer · AI Application Developer

kkuldangi2@gmail.com [GitHub ↗](#) [Blog ↗](#) [LinkedIn ↗](#)

SUMMARY

PDF conversion pass rate: 45.17% → 89.89%

RAG evidence coverage: 66.25% → 91.25%

Drug description conversion: \$200 estimated → \$11.18 spent

PROJECTS & EXPERIENCE

Fruition

2026.04 — Present

LLM Wiki-based AI workspace · AI SW Maestro, 17th cohort

Team project / Lead for all AI features

Teammates built the frontend and backend. I proposed the initial MSA design and finalized service boundaries, data ownership and communication with teammates and mentors.

[GitHub · local-pilot ↗](#) [GitHub · AI repository ↗](#)

Document conversion pipeline and faster Wiki generation

Converted uploaded PDFs for retrieval and grounded answers, and reduced Wiki generation time. Internal pass rate on 30 pages improved from 45.17% to 89.89%; analysis of 4 documents fell from 282.11s to 73.89s.

- PDF → Markdown conversion
 - Problem:** Open-source conversion damaged tables and equations; full AI conversion still had errors and high API usage.
 - Solution:** Compared PDF tools and adopted AnyDoc. Repaired damaged text with AI, processed tables and equations separately, and preserved original figures.
 - Result:** Internal pass rate on 30 PDF pages improved from 45.17% to 89.89%.
- Parallel Wiki generation
 - Problem:** Later documents waited for earlier external AI responses.
 - Solution:** Compared worker counts and requests per worker. Capped workers at 4 to manage memory; rejected extra concurrency and GPU parallelism that slowed processing.
 - Result:** Analysis of 4 documents fell from 282.11s to 73.89s (about 74% less).

Improving retrieval for grounded AI answers

Retrieved document evidence for AI answers. Coverage on 80 answerable questions rose from 66.25% to 91.25%, while false evidence returns on 20 unanswerable questions fell from 20 to 0.

- Evidence selection for RAG
 - Problem:** Retrieval missed required facts and returned evidence for unanswerable questions.
 - Solution:** Applied Jev after routing tests, then controlled candidates and return limits after a full-corpus trial. Tuned to 300 candidates and up to 4 parallel batches.
 - Result:** Across 9 papers, coverage on 80 answerable questions rose from 66.25% to 91.25%. False evidence returns on 20 unanswerable questions fell from 20 to 0. Median time before/after Jev tuning: 9.090s → 2.251s.

Reusable Skills and improved AI Agent routing

Built reusable Skills with user approval before changes. Applied Jev to route requests to questions, edits and other tasks: correct decisions rose from 77/98 to 81/98, and median decision time fell from 8.967s to 0.658s.

- User-defined task instructions (Skills)
 - Problem:** Developers cannot build every recurring workflow users need.
 - Solution:** Built drafting, review and publishing of natural-language instructions, with checks against bypassing permissions or approval. Document changes require separate preview and approval.
 - Result:** Published Skills became available to the Agent. In a 6-document check, no changes occurred before approval; 3 folders were created and documents moved afterward.
- Jev-based request routing
 - Problem:** Misclassifying questions as edits led to the wrong action.
 - Solution:** Applied Jev to select task types and additional conditions for questions, edits, document creation and folder organization.
 - Result:** Fully correct decisions on the same 98 requests rose from 77 to 81. Median decision time fell from 8.967s to 0.658s.

Service responsibilities and communication

Defined responsibilities, data ownership and communication for services including AI. Proposed the initial design and finalized it with teammates and mentors.

- Proposed the initial MSA design, then jointly finalized service boundaries, data ownership and communication with teammates and mentors.

Internal evaluations. RAG timing compares Jev before/after tuning. Parallel document timing covers analysis only, excluding storage and post-processing.

See the detailed portfolio for the problem-solving process and experimental results.

[PDF conversion](#) [Parallel document processing](#) [Agent Skill design](#) [RAG evidence selection](#)

Pilltip

2025.03 — 2025.12

Personalized AI medication service · Pusan National University
Team project / Backend and AI application development

Teammates owned Android, questionnaires, user DB modeling, login and other user features, and security including AES-GCM. The features below are my work.

[GitHub repository](#)

Drug data preparation pipeline

Rewrote about 44,000 drug records in plain language. Deduplication reduced input from 1GB to 326MB; after filling missing information, conversion cost \$11.18 in actual API spending versus about \$200 estimated for the original text.

- Problem:** Converting all 44,000 drug records was estimated to cost about \$200 in API calls.
Solution: Removed exact duplicates, then semantically equivalent sentences. Filled the missing 5% with ingredient-specific precautions, converted via GPT Batch API and mapped results back to each drug.
Result: Conversion input shrank from 1GB to 326MB. Actual API spending was \$11.18 versus about \$200 estimated for all original text.

Estimated cost versus actual spending; preprocessing labor is excluded.

AI medication chatbot using health profiles

Found drug and medication information from natural-language symptoms. Evaluated health profiles internally, separately from information sent to external AI.

- Problem:** Everyday symptom language differed from drug indications, while sensitive health data needed to stay out of external AI inputs.
Solution: Used Weaviate semantic search and AI orchestration for drug discovery, medication risks and dosage. Evaluated pregnancy, current medications and underlying conditions in internal code.
Result: Implemented personalized symptom-based search while excluding those profile fields from external LLM inputs.

Backend feature implementation

Built server features for drug search, risk information, medication reminders and logs, and family profiles. Integrated with teammates responsible for the mobile app, users and security.

- Designed the drug DB and implemented drug search, autocomplete and Drug Utilization Review (DUR) risk information with Java and Spring Boot.
- Implemented FCM medication reminders and logs, deep-link friend invitations, and family profile switching.
- Shared API documentation, meeting notes and architecture decision records in Notion.

See the detailed portfolio for the problem-solving process and experimental results.

[Data pipeline case study](#) [Personalization case study](#)

Skills

Backend	Java · Spring Boot · Spring AI · Python Authentication and authorization (Spring Security, OAuth 2.0) Personal shopping project · Login Todo web app · OAuth
AI & Search	LangChain · LangGraph · RAG · LLM Evaluation · Jev Elasticsearch · Weaviate · BGE-M3 · BM25
Data & Infra	MySQL · PostgreSQL · Redis · Kafka Docker · Docker Compose · GitHub Actions

Activities & contributions

2026.06 — 2026.06

Rhwp open-source contributions

Investigated bugs, submitted fixes and addressed CI in a Rust HWP/HWPX project. Fixed textFlow attributes of images, tables and shapes being reset after saving.

[GitHub repository](#) [PR #1213](#) · [Merged](#) [PR #1351](#) · [Submitted](#)

2025.03 — 2026.01



APPTIVE, Pusan National University

Backend mentee and mentor. Led 6 mentoring and code-review sessions for 12 mentees. Helped revise the curriculum to strengthen HTTP, Servlet, REST API and database foundations.

APPTIVE Backend Mentor Merit Award · 2026.01.09

2025.11 / 2025.07



SK AI SUMMIT · K-ICT WEEK in Busan

Represented Pusan National University at the Pilltip booth, demonstrating the service and answering technical questions.

Education & certification

2022.02 — 2026.02
(Transfer)

Pusan National University

GPA 4.12 / 4.5

2019.03 — 2022.02
(Withdrew)

Daegu University

GPA 4.2 / 4.5

2025.09

Engineer Information Processing

Human Resources Development Service of Korea

Awards

2025.10



Capstone Design Gold Award

Software & AI division / College of Information and Biomedical Engineering, Pusan National University

2025.09



Busan DATA WEEK Top Excellence Award

Data utilization case competition / Busan Techno Park

2025.09



AI LAUNCH Career School Startup Hackathon Encouragement Award

KRYPTON X x Root Impact x Google.org

2025.08



SW University Digital Competition Sponsor Award

SW University Council